Ho Van Duong

AI & Data Engineer Intern

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CAREER OBJECTIVE

Highly motivated IT student specializing in Artificial Intelligence. Seeking an AI Engineer Internship to apply solid foundations in Machine Learning, Computer Vision, and NLP to real-world projects. Eager to contribute to the team's success while sharpening technical skills in a professional environment.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT) 2022 - 2027

Bachelor of Information Technology — Specialization in Artificial Intelligence (AI)

GPA: 3.3 / 4.0

TECHNICAL SKILLS

Languages: Python (Primary), Java, C++, SQL
AI & Data: Machine Learning, Deep Learning (Computer Vision, NLP), Edge AI, Data Analysis, RAG, Multi-Agent Systems
Backend & DevOps: RESTful APIs, SQL Server, Vector Databases (FAISS), Docker, FastAPI, Flask, Next.js, Git/GitHub
Tools & Frameworks: TensorFlow, PyTorch/Keras, OpenCV, Scikit-learn, Pandas, NumPy

ACADEMIC PROJECTS

Beverage Brand Logo Recognition & Detection Pipeline (https://github.com/starduong/DL_Drink_Logo)
Tech Stack: Python, TensorFlow/Keras, PyTorch, VGG16, Faster R-CNN ResNet50-FPN, Albumentations, Selenium, ImageHash.

- Developed a 6-class brand classifier (VGG16) and a multi-object logo detector (Faster R-CNN ResNet50-FPN) trained completely without pretrained weights on a self-curated dataset of 11,100 classification images and 2,980 detection images (5,995 instances).
- Built a multi-source crawling script using Selenium and DuckDuckGo API integrated with perceptual hashing (imagehash) for near-duplicate filtering, alongside a PyTorch custom training loop utilizing mixed-precision acceleration, gradient clipping, and advanced Albumentations.
- **Key Outcome:** Achieved **96.95% Test Accuracy/F1-score** on the VGG16 brand classification task and reached a high localization performance of **92.48% mAP@50** using Faster R-CNN on edge-case out-of-sample test assets.

Real-Time Sign Language Translation System (https://github.com/starduong/HCI-Sign-Language-Edge)
Tech Stack: Python, MediaPipe, GCN/LSTM, TensorFlow Lite, OpenCV, Raspberry Pi 4.

- Built a real-time pipeline utilizing MediaPipe to extract skeletal landmarks and map spatial-temporal sequence data to text.
- Optimized and deployed a GCN/RNN architecture via TFLite on Raspberry Pi 4, ensuring low-latency inference on the edge.
- **Key Outcome:** Evaluated on a dataset of 2,000 UK sign language words, achieving over **70% accuracy** across all words, with 100 core words exceeding **90% accuracy**.

NoteMind: AI-Powered Hybrid-Cloud RAG System (https://github.com/starduong/NoteMind-RAG-Agent)
Tech Stack: Python, FastAPI, Next.js, FAISS, PyTorch, BAAI/bge-m3, GPT-4o, Gemini 3.5 Flash, GPT-4o-mini.

- Architected a Hybrid-Cloud RAG platform featuring an advanced NLP Ingestion Pipeline with Semantic Chunking (BAAI/bge-m3) and percentile-based breakpoint segmentation to maintain strict contextual coherence within a FAISS vector database.
- Engineered an Intelligent Multi-Agent Orchestrator driving 4 core workflows (Contextual Chat, Deep Research, Automated Quiz, Learning Roadmap) through a Layered LLM Strategy: leveraging GPT-4o for Agent Planning/Supervision, Gemini 3.5 Flash for long-context synthesis and Structured JSON Extraction, and GPT-4o-mini for millisecond-scale utility processing and automated citations.
- **Key Outcome:** Delivered a low-latency AI workspace enabling millisecond-scale semantic retrieval and high-accuracy, citation-grounded responses across multi-source knowledge bases with zero structural breakdown.

AWARDS

Consolation Prize, The 1st Vietnam AI Olympiad (OlpAI’25)	12/2025
Second Prize – AI Contest | Posts and Telecommunications Institute of Technology	10/2025